

ml

Tanvir

(data + learning algorithm) $\rightarrow$ function

1 Supervised learning

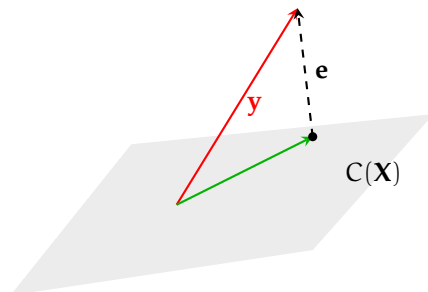
1.1 Regression

model, parameters, cost function, objective

1.1.1 Linear (in $\mathbf{w}$) Regression

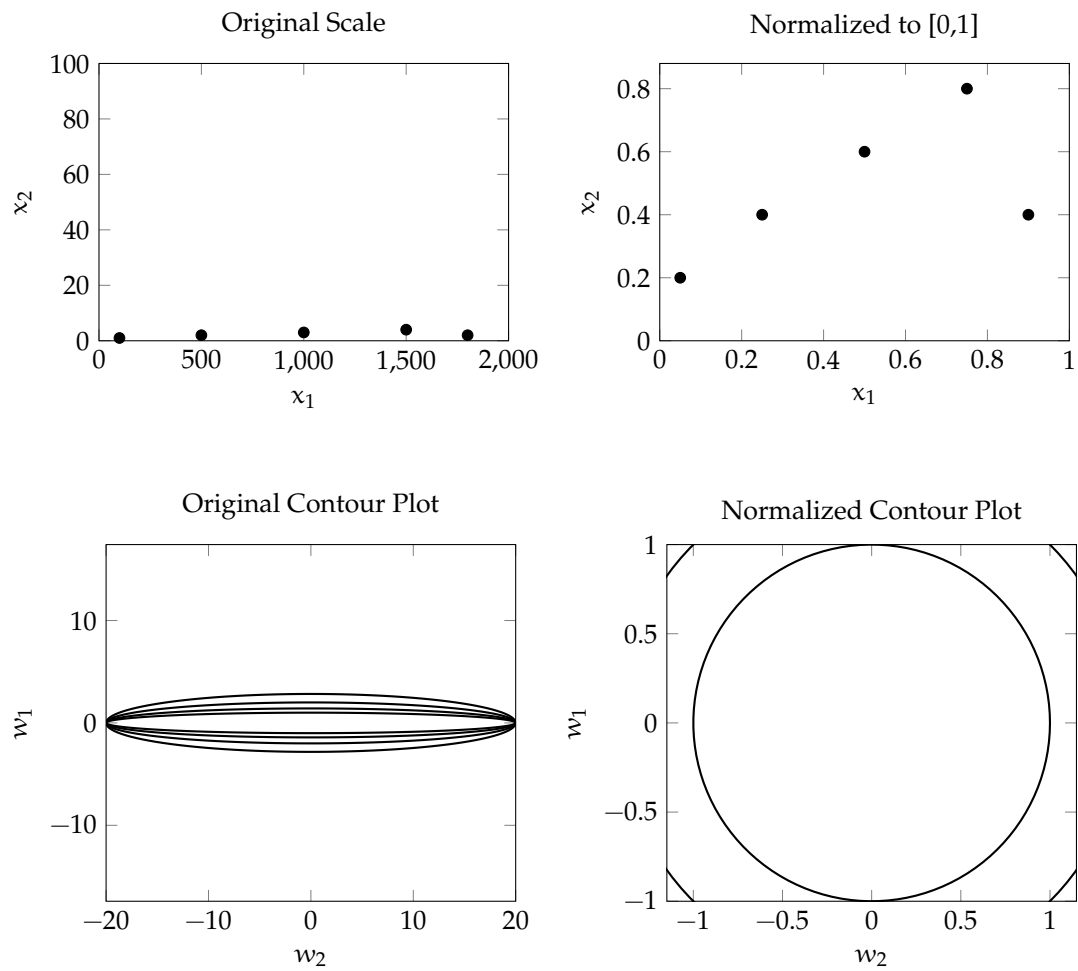
$\mathbf{X}^{m \times n}$ has m examples, each having n features. Usually $m \gg n$. $\mathbf{y}^{m \times 1}$ are the corresponding outputs.

$$\begin{aligned}\mathbf{X}\mathbf{w} &\approx \mathbf{y} \\ \text{error}(\mathbf{w}) &= \mathbf{X}\mathbf{w} - \mathbf{y} \\ \underset{\mathbf{w}}{\text{minimize}} \quad J(\mathbf{w}) &= \frac{1}{m} \|\text{error}(\mathbf{w})\|^2 = \frac{1}{m} (\mathbf{X}\mathbf{w} - \mathbf{y})^\top (\mathbf{X}\mathbf{w} - \mathbf{y}) \\ \frac{\partial J(\mathbf{w})}{\partial \mathbf{w}} &= \frac{2}{m} \mathbf{X}^\top (\mathbf{X}\mathbf{w} - \mathbf{y}) \\ \text{gradient descent: } \mathbf{w} &= \mathbf{w} - \alpha \frac{\partial J(\mathbf{w})}{\partial \mathbf{w}}\end{aligned}$$



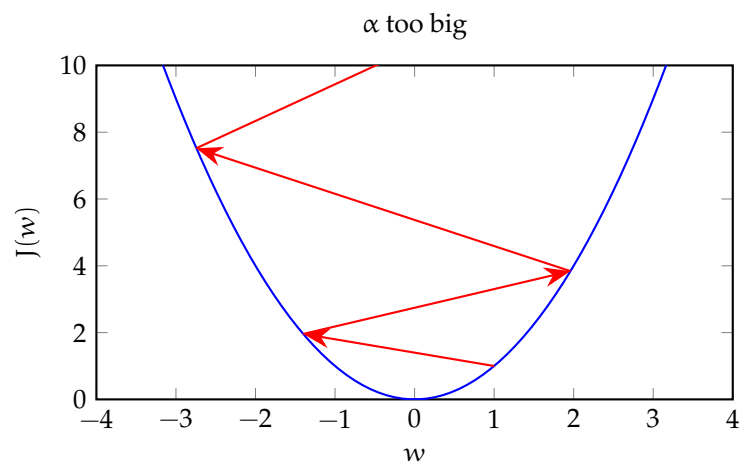
1.2 Normalization

Changes scale keeping the shape of distribution same. Gradient descent now works, with more ease, in the world of concentric circular contours.



1.3 Learning rate, α

A good value is somewhere between too big (divergence) and too small (slow convergence).



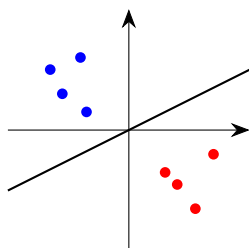
1.4 Classification

1.4.1 Logistic Regression

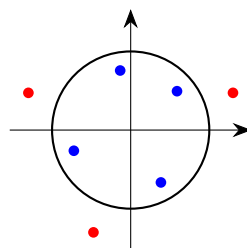
$$z = \mathbf{X}\mathbf{w}, \quad // \text{ linear regression}$$
$$\sigma(z) = \frac{1}{1 + e^{-z}} \quad // \text{ maps } z \text{ to probability-like } (0,1)$$

At $z = 0$, $\sigma(z) = 0.5$. So, with threshold 0.5, $z = 0$ is the decision boundary. As linear regression doesn't have to be straight line, logistic regression's decision boundary does not have to be a straight line.

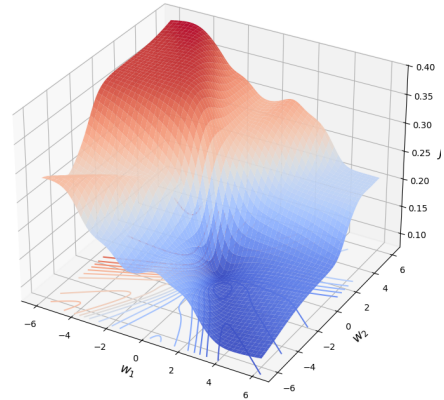
Linear boundary



Circular boundary



The squared error loss function $L(\mathbf{w}, y_i) = (\sigma(\mathbf{w}^T \mathbf{x}_i) - y_i)^2$, makes the cost function non-convex.

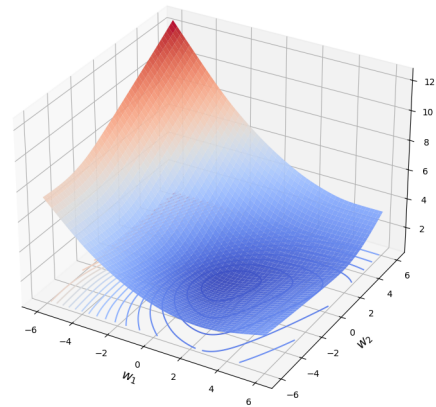


Negative log likelihood loss:

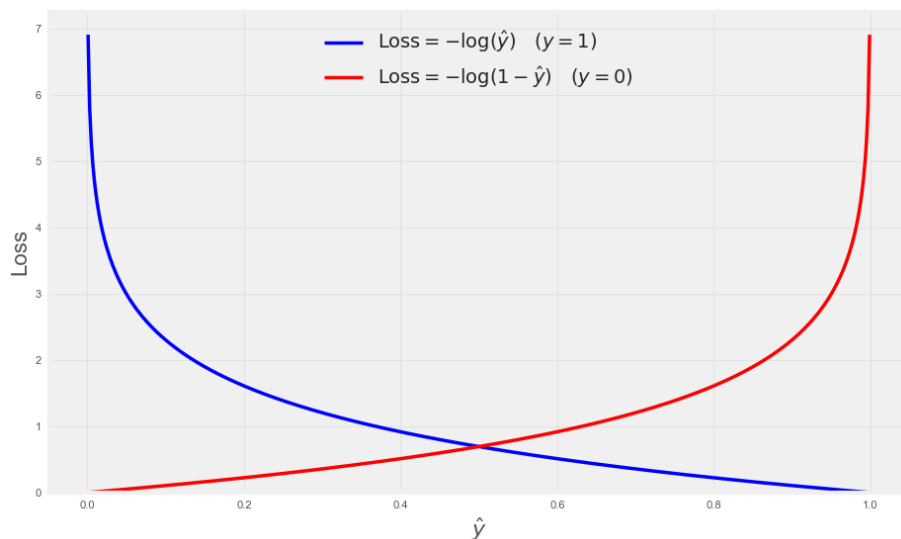
$$\hat{y}_i = \sigma(\mathbf{w}^T \mathbf{x}_i) \in (0, 1)$$

$$J(\mathbf{w}) = \frac{1}{m} \sum_{i=1}^m \begin{cases} -\log(\hat{y}_i), & \text{if } y_i = 1, \\ -\log(1 - \hat{y}_i), & \text{if } y_i = 0 \end{cases}$$

The negative log likelihood loss function with $\sigma(z)$ makes the cost function convex.



Negative log likelihood also incentivizes appropriately: big mismatch $\implies$ big loss, small mismatch $\implies$ small loss.



$$\mathbf{z}^{(i)} = \mathbf{w}^T \mathbf{x}^{(i)}$$

$$\text{Loss per example, } L(\mathbf{x}^{(i)}, \mathbf{w}) = -y^{(i)} \log(\sigma(\mathbf{z}^{(i)})) - (1 - y^{(i)}) \log(1 - \sigma(\mathbf{z}^{(i)}))$$

$$\frac{\partial \sigma(\mathbf{z}^{(i)})}{\partial \mathbf{w}} = \frac{\partial \sigma(\mathbf{z}^{(i)})}{\partial \mathbf{z}^{(i)}} \cdot \frac{\partial \mathbf{z}^{(i)}}{\partial \mathbf{w}} = \sigma(\mathbf{z}^{(i)}) (1 - \sigma(\mathbf{z}^{(i)})) \mathbf{x}^{(i)}$$

$$\frac{\partial L(\mathbf{x}^{(i)}, \mathbf{w})}{\partial \mathbf{w}} = (\sigma(\mathbf{z}^{(i)}) - y^{(i)}) \mathbf{x}^{(i)}$$

$$\frac{\partial J(\mathbf{X}, \mathbf{w})}{\partial \mathbf{w}} = \frac{1}{m} \sum_{i=1}^m \frac{\partial L(\mathbf{x}^{(i)}, \mathbf{w})}{\partial \mathbf{w}}$$

More concisely,

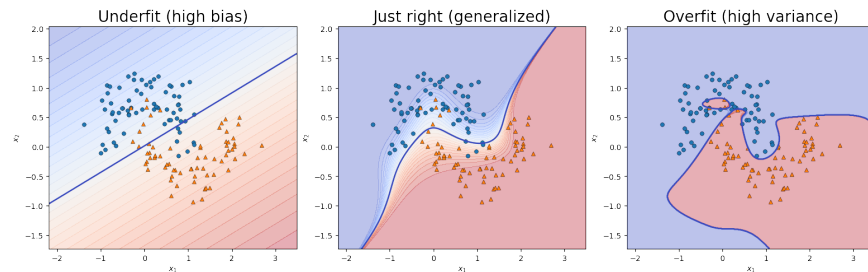
$$\mathbf{z} = \mathbf{X}\mathbf{w}$$

$$J(\mathbf{X}, \mathbf{w}) = -\frac{1}{m} [\mathbf{y}^T \log(\sigma(\mathbf{z})) + (\mathbf{1} - \mathbf{y})^T \log(1 - \sigma(\mathbf{z}))]$$

$$\frac{\partial J(\mathbf{X}, \mathbf{w})}{\partial \mathbf{w}} = \frac{1}{m} \mathbf{X}^T (\sigma(\mathbf{z}) - \mathbf{y})$$

$$\text{Gradient descent: } \mathbf{w} = \mathbf{w} - \alpha \frac{\partial J(\mathbf{X}, \mathbf{w})}{\partial \mathbf{w}}$$

2 Generalization



2.1 Address overfit

1. Collect more training examples keeping the model as it is.
2. Select a smaller set of informative features to simplify the model. Regularization disables features in a gradual manner.

2.1.1 Regularization

L2:

$$\begin{aligned} \underset{\mathbf{w}}{\text{minimize}} \quad J(\mathbf{w}) &= \frac{1}{m} \|\text{error}(\mathbf{w})\|^2 \\ \text{gradient descent:} \quad \mathbf{w} &= \mathbf{w} - \alpha \frac{\partial J(\mathbf{w})}{\partial \mathbf{w}} \\ \underset{\mathbf{w}}{\text{minimize}} \quad J_{\text{reg}}(\mathbf{w}) &= \frac{1}{m} \left(\|\text{error}(\mathbf{w})\|^2 + \lambda \|\mathbf{w}\|^2 \right) \\ \text{gradient descent (reg):} \quad \mathbf{w} &= \underbrace{\left(1 - \alpha \cdot \frac{\lambda}{m} \right)}_{\text{shrinkage}} \mathbf{w} - \alpha \frac{\partial J(\mathbf{w})}{\partial \mathbf{w}} \quad // \quad (\alpha \cdot \lambda/m) \sim 0 \end{aligned}$$

Here, $\lambda \|\mathbf{w}\|^2$ encourages smaller parameter values for less important features. With bigger λ 's, the parameters approach zero.

We can think regularization in terms of a constrained optimization problem:

$$\begin{aligned} \underset{\mathbf{w}}{\text{minimize}} \quad & J(\mathbf{w}) \\ \text{with constraint:} \quad & \|\mathbf{w}\|^2 \leq r^2 \end{aligned}$$

In other words, minimize $J(\mathbf{w})$ but do not let $\mathbf{w}$ go beyond a ball of radius r .

L1:

$$\underset{\mathbf{w}}{\text{minimize}} \quad J_{\text{reg}}(\mathbf{w}) = \frac{1}{m} \left(\|\text{error}(\mathbf{w})\|^2 + \lambda \|\mathbf{w}\|_1 \right)$$

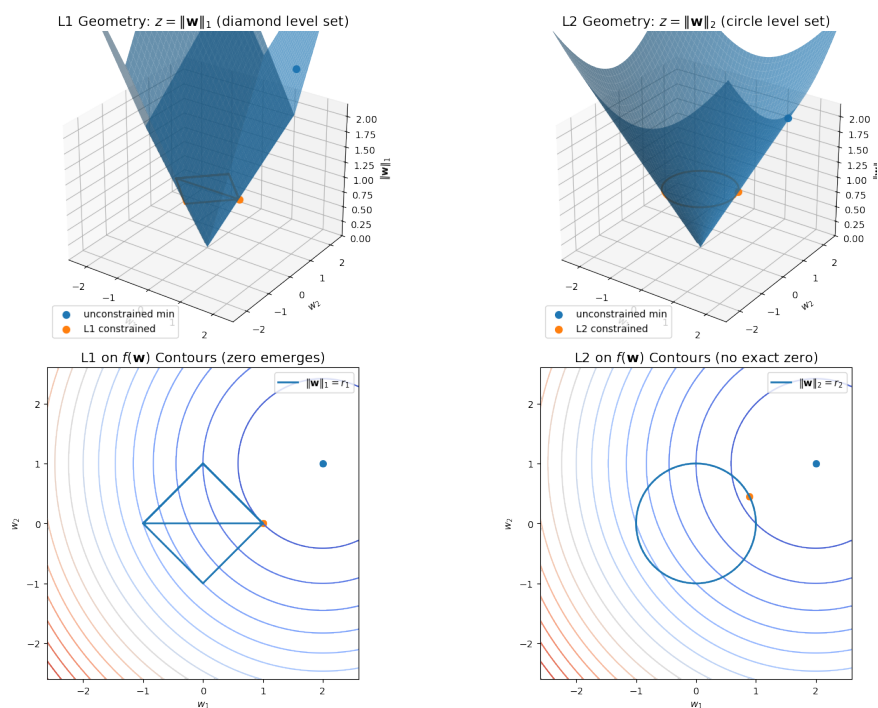
$$\|\mathbf{w}\|_1 = \sum_i |w_i|$$

$$\text{gradient descent (reg):} \quad \mathbf{w} = \mathbf{w} - \underbrace{\alpha \cdot \frac{\lambda}{m} \text{sign}(\mathbf{w})}_{\text{constant pull towards 0}} - \alpha \cdot \frac{\partial J(\mathbf{w})}{\partial \mathbf{w}}$$

The corresponding constrained optimization:

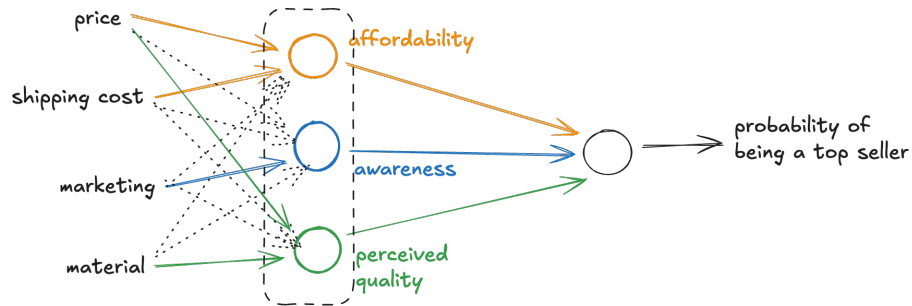
$$\underset{\mathbf{w}}{\text{minimize}} \quad J(\mathbf{w})$$

$$\text{with constraint:} \quad \|\mathbf{w}\|_1 \leq r$$



With L1 regularization, parameters for less important features can be exactly zero.

3 Neural Networks



- In multi-layer perceptron (MLP), an individual neuron can do linear or logistic regression. An MLP can be used both for regression and classification – the output layer needs adjusting.
- Intermediate layers of neurons pick out features in a hierarchical manner. Left to right, the features become more abstract – features from inputs, features of features, features of features of features, etc.
- Unlike other ML algorithms, a neural network's performance scales well with data.

4 References

- Machine learning specialization – Coursera